

Physics-Informed DMD (piDMD)

Aim: Learn structured linear models from data.

~~Recap~~ Given snapshot matrices

$$X = \begin{bmatrix} 1 & & 1 \\ x_1 & \dots & x_m \\ 1 & & 1 \end{bmatrix} \quad \text{and} \quad Y = \begin{bmatrix} 1 & & 1 \\ y_1 & \dots & y_m \\ 1 & & 1 \end{bmatrix},$$

$d \times m$ $d \times m$

where $y_j = \overline{T} x_j \quad j=1, \dots, m$

DMD models minimize the objective:

$$\hat{T}_{\text{DMD}} = \underset{S \in \mathbb{R}^{d \times d}}{\operatorname{argmin}} \|Y - SX\|_F,$$

and $\hat{T}_{\text{DMD}} = YX^\dagger$ is low-rank when $m \ll n$.

piDMD constrains S to a matrix manifold:

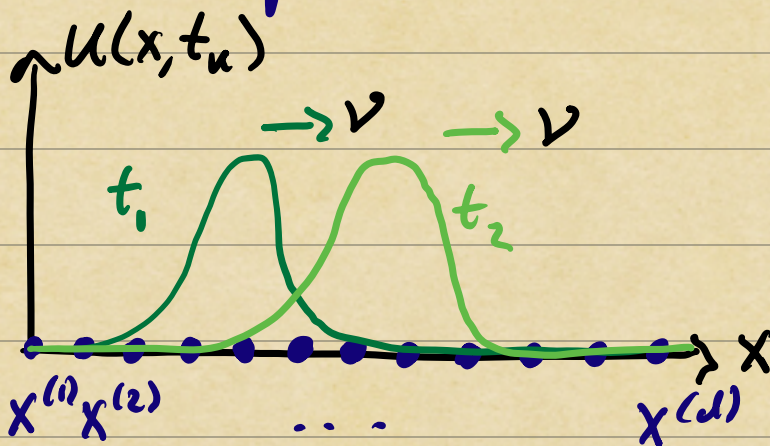
$$\hat{T}_{\text{pi}} = \underset{S \in M}{\operatorname{argmin}} \|Y - SX\|_F.$$

The manifold M encodes the physical properties of the dynamical system:

$$x_{k+1} = T x_k, \quad k=0, 1, 2, \dots$$

which constrain the entries/structure of T .

Example. 1D Advection



$$\begin{aligned} \partial_t u &= v \partial_x u \\ u(x, 0) &= g(x) \\ u(0, t) &= u(1, t) \end{aligned}$$

We can write down the exact soln. to the Advection PDE w/a Fourier Series approach.

$$v \partial_x v = \lambda v \quad \xrightarrow[\text{Periodic B.C.'s}]{} v_k = e^{2\pi i k x}, \quad \lambda_k = 2\pi i k v, \quad k=0, \pm 1, \dots$$

$$u(x, t) = \sum_{k=-\infty}^{+\infty} \hat{g}_k e^{2\pi i k (x + vt)} = g(x + vt)$$

$\hat{g}_k$ Fourier Coeffs of $g(x)$

Q: What physical/mathematical properties of the solution could you aim to preserve in your pDMP data-driven model?

1) **Time-periodic.** Solution repeats w/period $T = 1/v$,

$$u(x, t) = u(x, t + v^{-1}).$$

2) **Causal.** Information travels at finite-speed and, in particular, to the right ($v < 0$) or left ($v > 0$):

$$u(x, t) = g(x + vt) = u(x + v\tau, t - \tau),$$

provided that $x + v\tau < 1$ ($v > 0$) or $x + v\tau > 0$ ($v < 0$).

3) **Conservative.** The $L^2([0, 1])$ norm of the soln. is conserved for all $t \geq 0$:

$$\int_0^1 |u(x, t)|^2 dx = \int_0^1 |g(x)|^2 dx$$

4) **Shift Invariant.** Shifting the soln. is the

Same as shifting the initial condition. If $u(x,t) = \exp(tv\partial_x)g(x)$ and $[\mathcal{S}_y, \mathcal{S}](x) = \mathcal{S}(x+y)$,

$$\mathcal{S}_y u(x,t) = \exp(tv\partial_x) \mathcal{S}_y g(x), \quad 0 \leq y \leq 1.$$

The soln. operator commutes w/shifts.

In practice, we are trying to learn dynamics associated w/discrete spatial & temporal data:

$$u_{k+1} = \overline{T} u_k, \quad k = 0, 1, 2, \dots$$

where $u_k^{(j)} = u(j\Delta x, k\Delta t)$ for $\Delta x, \Delta t \geq 0$.

Q: How do the physical properties above manifest in the matrix $\overline{T}$?

It can be helpful to look at finite-difference discretizations of the PDE to see how physical properties correspond to algebraic structure.

One possible discrete model on the grid:

$$\frac{d}{dt} \begin{bmatrix} u^{(1)} \\ \vdots \\ u^{(d)} \end{bmatrix} \approx \frac{1}{2\Delta x} \begin{bmatrix} 0 & 1 & \cdots & -1 \\ -1 & 0 & 1 & \cdots \\ \vdots & \ddots & \ddots & \vdots \\ 1 & \cdots & -1 & 0 \end{bmatrix} \begin{bmatrix} u^{(1)} \\ \vdots \\ u^{(d)} \end{bmatrix}$$

$\underbrace{\quad}_{T_1} \quad A_1$

$$u_{k+1} = \exp(\underbrace{A_1}_{T_1} \Delta t) u_k, \quad k=0,1,2,\dots$$

This is derived from $\partial_x u|_{x=x_k} \approx \frac{u(x_k+h) - u(x_k-h)}{2h}$.

A second possible model on the grid ($\nu < 0$)

$$\frac{d}{dt} \begin{bmatrix} u^{(1)} \\ \vdots \\ u^{(d)} \end{bmatrix} \approx \frac{1}{2\Delta x} \begin{bmatrix} 1 & \cdots & -1 \\ -1 & 1 & \cdots \\ \vdots & \ddots & \ddots \\ -1 & \cdots & 1 \end{bmatrix} \begin{bmatrix} u^{(1)} \\ \vdots \\ u^{(d)} \end{bmatrix}$$

$\underbrace{\quad}_{T_2} \quad A_2$

$$u_{k+1} = \exp(\underbrace{A_2}_{T_2} \Delta t) u_k, \quad k=0,1,2,\dots$$

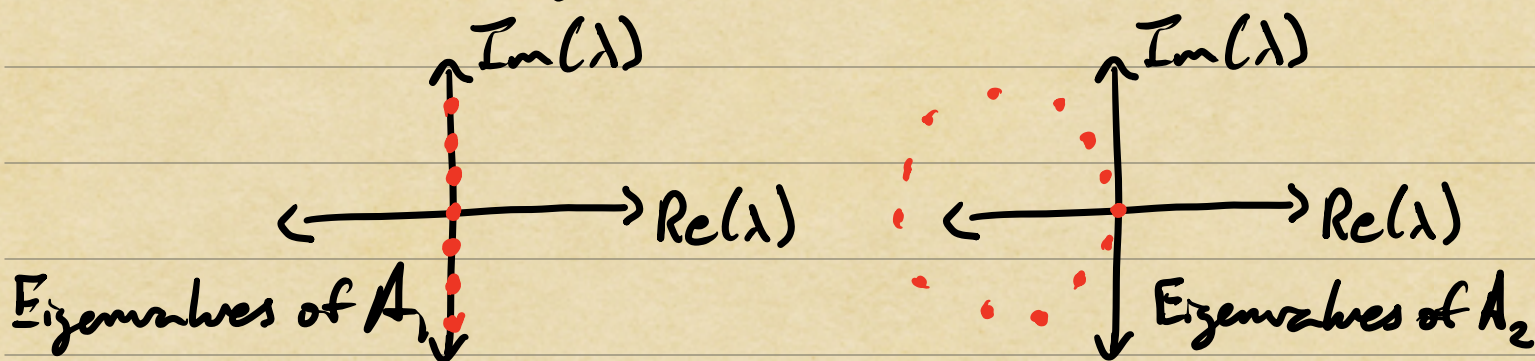
This is derived from $\partial_x u|_{x=x_k} \approx \frac{u(x_k) - u(x_k-h)}{h}$

Q: Which discretization preserves which physical properties of the PDE soln?

$\Rightarrow$ The solutions of the discrete dynamical systems are time-periodic if the eigenvalues of T lie on the unit circle - equivalently, if the eigenvalues of A are imaginary.

$$\lambda(A) \subset i\mathbb{R} \iff |\lambda(T)| = |\exp(\lambda(A))| = 1$$

The matrix A_1 is anti-symmetric ($A_1^T = -A_1$) and, therefore, has imaginary eigenvalues. A_2 has eigenvalues with $\text{Im}(\lambda) < 0$.



So A_1 preserves periodicity, but A_2 doesn't.

$\Rightarrow$ The solutions of the discrete dynamical system are causal if (say, for $\nu < 0$)

$$u_{k+1}^{(i)} = [\exp(A\Delta t)u_k]^{(i)} = \sum_{j=1}^i c_j u_k^{(j)},$$

for some coefficients c_i . This occurs only if $\exp(A\Delta t)$ is lower triangular, which occurs if A itself is lower triangular, since

$$\exp(A\Delta t) = \sum_{j=0}^{\infty} \frac{1}{j!} (\Delta t A)^j$$

and each term remains triangular if A is.

The matrix A_2 is lower triangular except for the first row, due to the periodic b.c. linking $x=0$ and $x=1$. Therefore, A_2 enforces local causality away from the periodic boundary in the discrete model.

The matrix A_1 uses information from the left and right (super- and sub-diagonal) to update the discrete system, so it does not respect local causality.

$\Rightarrow$ The discrete dynamical system is conservative (preserves the discrete 2-norm)

if $\exp(A\Delta t)$ is unitary/orthogonal. This happens precisely, when A has imaginary eigenvalues and an orthogonal basis of eigenvectors, i.e., when A is anti-Hermitian. Therefore, A_1 respects the conservative property and A_2 doesn't.

$\Rightarrow$ The discrete dynamical system is shift invariant if $\exp(A\Delta t)$ commutes with the discrete shift operator, which happens if A commutes, since

$$\exp(A\Delta t) = \sum_{j=0}^{\infty} \frac{1}{j!} (\Delta t A)^j.$$

For a periodic domain, this implies that A must be circulant, i.e., constant along diagonals, with diagonals "wrapping" around at the end of rows.

Circulant
matrix

$$C = \begin{bmatrix} c_0 & c_1 & c_2 & \dots & c_{n-1} \\ & c_0 & c_1 & \dots & c_{n-1} \\ & & \ddots & \ddots & \ddots \\ c_2 & \dots & \dots & c_1 & c_0 \\ c_1 & c_2 & \dots & c_0 & c_1 \end{bmatrix}$$

Both A_1 and A_2 are circulant so both discrete dynamical systems respect the shift-invariant property.

Q: How do the physical properties above manifest in the matrix T ? In A ?

Collecting our observations about the discrete models into a table yields these equivalences:

Physical Properties

Matrix Structure

Time-periodic

$\iff$

Eigenvals of A imaginary
" T on unit circle

Shift-Invariant

$\iff$

A, T circulant
(periodic grid)

Conservative

$\iff$

A anti-Hermitian
 T unitary

Causal

$\iff$

A, T triangular